

ForEveryOne

File

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CALCULATION OF A HELICAL GEAR PAIR

Drawing or article number:

Gear 1: 0.000.0

Gear 2: 0.000.0

Calculation method ISO 6336:2006 Method B

----- GEAR 1 ----- GEAR 2 --

Power (kW)	[P]	3.000	
Speed (1/min)	[n]	750.0	228.7
Torque (Nm)	[T]	38.2	125.3
Application factor	[KA]	1.25	
Required service life (h)	[H]	31000.00	
Gear driving (+) / driven (-)		+	-
Working flank gear 1: Right flank			
Sense of rotation gear 1 clockwise			

1. TOOTH GEOMETRY AND MATERIAL

(geometry calculation according to ISO 21771:2007, DIN ISO 21771)

----- GEAR 1 ----- GEAR 2 --

Center distance (mm)	[a]	178.268	
Center distance tolerance	ISO 286:2010 Measure js7		
Normal module (mm)	[mn]	3.0000	
Pressure angle at normal section (°)	[alfn]	20.0000	
Helix angle at reference circle (°)	[beta]	25.0000	
Number of teeth	[z]	25	82
Facewidth (mm)	[b]	56.30	54.80
Hand of gear		right	left
Accuracy grade	[Q-ISO 1328:1995]	6	6
Inner diameter (mm)	[di]	0.00	0.00
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00

Material

Gear 1: C45 (1), Through hardened steel, unalloyed, through hardened
ISO 6336-5 Figure 5/6 (MQ)

Gear 2: C45 (1), Through hardened steel, unalloyed, through hardened
ISO 6336-5 Figure 5/6 (MQ)

----- GEAR 1 ----- GEAR 2 --

Surface hardness		HBW 186	HBW 186
Material quality according to ISO 6336:2006 Normal (Life factors ZNT and YNT >=0.85)			
Fatigue strength. tooth root stress (N/mm²)	[σFlim]	210.00	210.00
Fatigue strength for Hertzian pressure (N/mm²)	[σHlim]	540.00	540.00
Tensile strength (N/mm²)	[σB]	700.00	700.00
Yield point (N/mm²)	[σS]	490.00	490.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[ν]	0.300	0.300
Roughness average value DS, flank (μm)	[RAH]	1.05	1.05

Roughness average value DS, root (µm)	[RAF]	3.00	3.00
Mean roughness height, Rz, flank (µm)	[RZH]	8.00	8.00
Mean roughness height, Rz, root (µm)	[RZF]	20.00	20.00

Gear reference profile 1 :

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A		
Dedendum coefficient	[hfP*]	1.250	
Root radius factor	[rhofP*]	0.380 (rhofPmax*=0.472)	
Addendum coefficient	[haP*]	1.000	
Tip radius factor	[rhoaP*]	0.000	
Protuberance height coefficient	[hprP*]	0.000	
Protuberance angle	[alfprP]	0.000	
Tip form height coefficient	[hFaP*]	0.000	
Ramp angle	[alfKP]	0.000	
	not topping		

Gear reference profile 2 :

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A		
Dedendum coefficient	[hfP*]	1.250	
Root radius factor	[rhofP*]	0.380 (rhofPmax*=0.472)	
Addendum coefficient	[haP*]	1.000	
Tip radius factor	[rhoaP*]	0.000	
Protuberance height coefficient	[hprP*]	0.000	
Protuberance angle	[alfprP]	0.000	
Tip form height coefficient	[hFaP*]	0.000	
Ramp angle	[alfKP]	0.000	
	not topping		

Summary of reference profile gears:

Dedendum reference profile	[hfP*]	1.250	1.250
Tooth root radius Refer. profile	[rofP*]	0.380	0.380
Addendum Reference profile	[haP*]	1.000	1.000
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[alfprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[alfKP]	0.000	0.000

Type of profile modification: none (only running-in)

Tip relief (µm)	[Ca]	10.7	10.7
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Lubrication type Oil bath lubrication

Type of oil Oil: ISO-VG 220

Lubricant base Mineral-oil base

Kinem. viscosity oil at 40 °C (mm²/s)	[nu40]	220.00
Kinem. viscosity oil at 100 °C (mm²/s)	[nu100]	17.50
Specific density at 15 °C (kg/dm³)	[roOil]	0.895
Oil temperature (°C)	[TS]	70.000

----- GEAR 1 ----- GEAR 2 --

Overall transmission ratio	[itot]	-3.280
Gear ratio	[u]	3.280
Transverse module (mm)	[mt]	3.310
Pressure angle at pitch circle (°)	[alf]	21.880
Working transverse pressure angle (°)	[alfwt]	22.803
	[alfwt.e/i]	22.818 / 22.788
Working pressure angle at normal section (°)	[alfwn]	20.836
Helix angle at operating pitch circle (°)	[betaw]	25.146

Base helix angle (°)	[betab]	23.399		
Reference center distance (mm)	[ad]	177.092		
Sum of profile shift coefficients	[Summexi]	0.4000		
Profile shift coefficient	[x]	0.3013	0.0987	
Tooth thickness (Arc) (module) (module)	[sn*]	1.7901	1.6426	
Tip alteration (mm)	[k*mn]	-0.024	-0.024	
Reference diameter (mm)	[d]	82.753	271.431	
Base diameter (mm)	[db]	76.792	251.878	
Tip diameter (mm)	[da]	90.513	277.975	
(mm)	[da.e/i]	90.513 / 90.478	277.975 / 277.923	
Tip diameter allowances (mm)	[Ada.e/i]	0.000 / -0.035	0.000 / -0.052	
Tip form diameter (mm)	[dFa]	90.513	277.975	
(mm)	[dFa.e/i]	90.513 / 90.478	277.975 / 277.923	
Active tip diameter (mm)	[dNa]	90.513	277.975	
Active tip diameter (mm)	[dNa.e/i]	90.513 / 90.478	277.975 / 277.923	
Operating pitch diameter (mm)	[dw]	83.303	273.234	
(mm)	[dw.e/i]	83.312 / 83.294	273.264 / 273.203	
Root diameter (mm)	[df]	77.061	264.523	
Generating Profile shift coefficient	[xE.e/i]	0.2693/ 0.2509	0.0552/ 0.0323	
Generated root diameter with xE (mm)	[df.e/i]	76.869 / 76.759	264.262 / 264.125	
Theoretical tip clearance (mm)	[c]	0.750	0.750	
Effective tip clearance (mm)	[c.e/i]	0.987 / 0.861	0.947 / 0.826	
Active root diameter (mm)	[dNf]	79.505	267.565	
(mm)	[dNf.e/i]	79.564 / 79.478	267.623 / 267.531	
Root form diameter (mm)	[dFf]	79.252	266.364	
(mm)	[dFf.e/i]	79.126 / 79.055	266.137 / 266.018	
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.254 / 0.176	0.802 / 0.697	
Addendum (mm)	[ha=mn*(haP*+x+k)]	3.880	3.272	
(mm)	[ha.e/i]	3.880 / 3.862	3.272 / 3.246	
Dedendum (mm)	[hf=mn*(hfP*-x)]	2.846	3.454	
(mm)	[hf.e/i]	2.942 / 2.997	3.584 / 3.653	
Roll angle at dFa (°)	[xsi_dFa.e/i]	35.748 / 35.699	26.749 / 26.721	
Roll angle to dNa (°)	[xsi_dNa.e/i]	35.748 / 35.699	26.749 / 26.721	
Roll angle to dNf (°)	[xsi_dNf.e/i]	15.532 / 15.287	20.572 / 20.510	
Roll angle at dFf (°)	[xsi_dFf.e/i]	14.232 / 14.012	19.550 / 19.466	
Tooth height (mm)	[h]	6.726	6.726	
Virtual gear no. of teeth	[zn]	32.749	107.418	
Normal tooth thickness at tip circle (mm)	[san]	2.034	2.427	
(mm)	[san.e/i]	1.977 / 1.915	2.351 / 2.279	
Normal tooth thickness on tip form circle (mm)	[sFan]	2.034	2.427	
(mm)	[sFan.e/i]	1.977 / 1.915	2.351 / 2.279	
Normal space width at root circle (mm)	[efn]	2.441	2.153	
(mm)	[efn.e/i]	2.490 / 0.000	2.168 / 2.176	
Max. sliding velocity at tip (m/s)	[vga]	0.801	0.599	
Specific sliding at the tip	[zetaa]	0.426	0.426	
Specific sliding at the root	[zetaf]	-0.741	-0.741	
Mean specific sliding	[zetam]	0.426		
Sliding factor on tip	[Kga]	0.245	0.183	
Sliding factor on root	[Kgf]	-0.183	-0.245	
Pitch on reference circle (mm)	[pt]	10.399		
Base pitch (mm)	[pbt]	9.650		
Transverse pitch on contact-path (mm)	[pet]	9.650		
Lead height (mm)	[pz]	557.523	1828.676	
Axial pitch (mm)	[px]	22.301		
Length of path of contact (mm)	[ga, e/i]	13.660 (13.712 / 13.514)		
Length T1-A, T2-A (mm)	[T1A, T2A]	10.296(10.244/ 10.409)	58.795(58.795/ 58.733)	
Length T1-B (mm)	[T1B, T2B]	14.306(14.306/ 14.273)	54.785(54.733/ 54.869)	

Length T1-C (mm)	[T1C, T2C]	16.143(	16.131/	16.155)	52.948(	52.909/	52.988)
Length T1-D (mm)	[T1D, T2D]	19.946(	19.894/	20.059)	49.145(	49.145/	49.083)
Length T1-E (mm)	[T1E, T2E]	23.956(	23.956/	23.923)	45.135(	45.083/	45.219)
Length T1-T2 (mm)	[T1T2]	69.091 (69.039 / 69.142)					
Diameter of single contact point B (mm)	[d-B]	81.949(	81.949/	81.926)	274.678(	274.637/	274.746)
Diameter of single contact point D (mm)	[d-D]	86.535(	86.488/	86.640)	270.377(	270.377/	270.332)
Addendum contact ratio	[eps]	0.810(	0.811/	0.805)	0.606(	0.610/	0.595)
Minimal length of contact line (mm)	[Lmin]	79.908					
Transverse contact ratio	[eps_a]	1.416					
Transverse contact ratio with allowances	[eps_a.e/m/i]	1.421 / 1.411 / 1.400					
Overlap ratio	[eps_b]	2.457					
Total contact ratio	[eps_g]	3.873					
Total contact ratio with allowances	[eps_g.e/m/i]	3.878 / 3.868 / 3.858					

2. FACTORS OF GENERAL INFLUENCE

		----- GEAR 1 -----	GEAR 2 --
Nominal circum. force at pitch circle (N)	[Ft]		923.2
Axial force (N)	[Fa]		430.5
Radial force (N)	[Fr]		370.7
Normal force (N)	[Fnorm]		1084.0
Nominal circumferential force per mm (N/mm)	[w]		16.85
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]		917.1
Axial force (N)	[Faw]		430.5
Radial force (N)	[Frw]		385.6
Circumferential speed reference circle (m/s)	[v]		3.25
Circumferential speed operating pitch circle (m/s)	[v(dw)]		3.27
Running-in value (μm)	[yp]		2.5
Running-in value (μm)	[yf]		2.8
Correction factor	[CM]		0.800
Gear blank factor	[CR]		1.000
Basic rack factor	[CBS]		0.975
Material coefficient	[E/Est]		1.000
Singular tooth stiffness (N/mm/μm)	[c']		9.710
Meshing stiffness (N/mm/μm)	[cgalf]		12.737
Meshing stiffness (N/mm/μm)	[cgbet]		10.826
The formula for c' and cg at w*KA < 25 N/mm is imprecise!			
c', cg is calculated with w*KA = 25 N/mm.			
Reduced mass (kg/mm)	[mRed]		0.02341
Resonance speed (min-1)	[nE1]		8910
Resonance ratio (-)	[N]		0.084
Subcritical range			
Running-in value (μm)	[ya]		2.5
Bearing distance l of pinion shaft (mm)	[l]		112.600
Distance s of pinion shaft (mm)	[s]		11.260
Outside diameter of pinion shaft (mm)	[dsh]		56.300
Load in accordance with Figure 13, ISO 6336-1:2006	[-]	4	
0:a), 1:b), 2:c), 3:d), 4:e)			
Coefficient K' according to Figure 13, ISO 6336-1:2006	[K']	-1.00	
Without stiffening			
Tooth trace deviation (active) (μm)	[Fby]		3.00
from deformation of shaft (μm)	[fsh*B1]		0.11
(fsh (μm) = 0.11, B1= 1.00, fHb5 (μm) = 7.50)			
Tooth without tooth trace modification			

Position of Contact pattern: favorable from production tolerances (μm) ($B2= 1.00$)	[fma*B2]	14.14	
Tooth trace deviation, theoretical (μm)	[Fbx]	7.35	
Running-in value (μm)	[yb]	4.36	
Dynamic factor	[KV]	1.118	
Face load factor - flank	[KHb]	1.689	
- Tooth root	[KFb]	1.583	
- Scuffing	[KBb]	1.689	
Transverse load factor - flank	[KHa]	1.994	
- Tooth root	[KF _a]	1.994	
- Scuffing	[KB _a]	1.994	
Helical load factor scuffing	[Kbg]	1.300	
Number of load cycles (in mio.)	[NL]	1395.000	425.305

3. TOOTH ROOT STRENGTH

Calculation of Tooth form coefficients according method: B

		----- GEAR 1 -----	GEAR 2 --
Calculated with profile shift	[x]	0.3013	0.0987
Tooth form factor	[YF]	1.12	1.22
Stress correction factor	[YS]	2.27	2.24
Load application angle (°)	[alfFen]	20.94	20.25
Bending moment arm (mm)	[hF]	2.72	3.16
Tooth thickness at root (mm)	[sFn]	6.60	6.83
Tooth root radius (mm)	[roF]	1.38	1.34
(hF* = 0.907/ 1.053 sFn* = 2.199/ 2.278 roF* = 0.458/ 0.446)			
(den (mm) = 100.633/324.292 dsFn(mm) = 78.194/265.709 alfsFn(°) = 30.00/ 30.00 qs = 2.399/ 2.554)			
Helix angle factor	[Ybet]	0.792	
Deep tooth factor	[YDT]	1.000	
Gear rim factor	[YB]	1.00	1.00
Effective facewidth (mm)	[beff]	56.30	54.80
Nominal stress at tooth root (N/mm²)	[sigF0]	11.00	12.09
Tooth root stress (N/mm²)	[sigF]	48.51	53.35
Permissible bending stress at root of Test-gear			
Notch sensitivity factor	[YdreIT]	0.997	1.001
Surface factor	[YRrelT]	0.957	0.957
size factor (Tooth root)	[YX]	1.000	1.000
Finite life factor	[YNT]	0.884	0.906
	[YdreIT*YRrelT*YX*YNT]	0.844	0.868
Alternating bending factor (mean stress influence coefficient)	[YM]	1.000	1.000
Stress correction factor	[Yst]	2.00	
Yst*sigFlim (N/mm²)	[sigFE]	420.00	420.00
Permissible tooth root stress (N/mm²)	[sigFP=sigFG/SFmin]	253.11	260.27
Limit strength tooth root (N/mm²)	[sigFG]	354.35	364.38
Required safety	[SFmin]	1.40	1.40
Safety for tooth root stress	[SF=sigFG/sigF]	7.30	6.83
Transmittable power (kW)	[kWRating]	15.65	14.64

4. SAFETY AGAINST PITTING (TOOTH FLANK)

		----- GEAR 1 -----	GEAR 2 --
Zone factor	[ZH]		2.252
Elasticity factor ($\sqrt{N/mm^2}$)	[ZE]		189.812
Contact ratio factor	[Zeps]		0.840
Helix angle factor	[Zbet]		1.050
Effective facewidth (mm)	[beff]		54.80
Nominal contact stress (N/mm ²)	[sigH0]		194.47
Contact stress at operating pitch circle (N/mm ²)	[sigHw]		421.93
Single tooth contact factor	[ZB,ZD]	1.00	1.00
Contact stress (N/mm ²)	[sigHB, sigHD]	421.93	421.93
Lubrication coefficient at NL	[ZL]	1.038	1.038
Speed coefficient at NL	[ZV]	0.942	0.942
Roughness coefficient at NL	[ZR]	0.872	0.872
Material pairing coefficient at NL	[ZW]	1.000	1.000
Finite life factor	[ZNT]	0.903	0.936
	[ZL*ZV*ZR*ZNT]	0.770	0.799
Limited pitting is permitted:	No		
Size factor (flank)	[ZX]	1.000	1.000
Permissible contact stress (N/mm ²)	[sigHP=sigHG/SHmin]	415.82	431.25
Pitting stress limit (N/mm ²)	[sigHG]	415.82	431.25
Required safety	[SHmin]	1.00	1.00
Safety factor for contact stress at operating pitch circle			
	[SHw]	0.99	1.02
Safety for stress at single tooth contact	[SHBD=sigHG/sigHBD]	0.99	1.02
(Safety regarding transmittable torque)	[(SHBD)^2]	0.97	1.04
Transmittable power (kW)	[kWRating]	2.91	3.13

4b. MICROPITTING ACCORDING TO ISO/TR 15144-1:2014

Calculation did not run. (Lubricant: Load stage micropitting test is unknown.)

5. SCUFFING LOAD CAPACITY

Calculation method according to ISO/TS 6336-20/21:2017

Lubrication coefficient (for lubrication type)	[XS]	1.000	
Scuffing test and load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1)	12
Multiple meshing factor	[Xmp]	1.000	
Relative structure coefficient (Scuffing)	[XWrelT]	1.000	
Thermal contact factor (N/mm/s ^{0.5} /K)	[BM]	13.780	13.780
Relevant tip relief (μm)	[Ca]	10.70	10.70
Optimal tip relief (μm)	[CeFF]	1.65	
Ca taken as optimal in the calculation (0=no, 1=yes)		0	0
Effective facewidth (mm)	[beff]	54.800	
Applicable circumferential force/facewidth (N/mm)	[wBt]	79.296	
Kbg = 1.300, wBt*Kbg = 103.085			
Angle factor	[Xalfbet]	1.000	
(ε1:0.810, ε2:0.606)			

Flash temperature-criteria		
Lubricant factor	[XL]	0.830
Tooth mass temperature (°C) (theMi = theoil + XS*0.47*Xmp*theflm)	[theMi]	71.38
Average flash temperature (°C)	[theflm]	2.94
Scuffing temperature (°C)	[theS]	348.80
Coordinate gamma (point of highest temp.) [Gamma.A]=-0.362 [Gamma.E]=0.484	[Gamma]	0.484
Highest contact temp. (°C)	[theB]	77.55
Flash factor (°K*N ^{-1.75} *s ^{-1.5} *m ^{-1.5} mm)	[XM]	50.058
Approach factor	[XJ]	1.000
Load sharing factor	[XGam]	0.706
Dynamic viscosity (mPa*s)	[etaM]	41.90 (70.0 °C)
Coefficient of friction	[mym]	0.053
Required safety	[SBmin]	2.000
Safety factor for scuffing (flash temperature)	[SB]	36.924
Integral temperature-criteria		
Lubricant factor	[XL]	1.000
Tooth mass temperature (°C) (theMC = theoil + XS*0.70*theflaint)	[theMC]	71.09
Mean flash temperature (°C)	[theflaint]	1.56
Integral scuffing temperature (°C)	[theSint]	360.78
Flash factor (°K*N ^{-1.75} *s ^{-1.5} *m ^{-1.5} mm)	[XM]	50.058
Running-in factor (well run in)	[XE]	1.000
Contact ratio factor	[Xeps]	0.275
Dynamic viscosity (mPa*s)	[etaOil]	41.90 (70.0 °C)
Mean coefficient of friction	[mym]	0.054
Geometry factor	[XBE]	0.218
Meshing factor	[XQ]	1.000
Tip relief factor	[XCa]	1.660
Integral tooth flank temperature (°C)	[theint]	73.43
Required safety	[SSmin]	1.800
Safety factor for scuffing (intg.-temp.)	[SSint]	4.913
Safety referring to transmittable torque	[SSL]	84.686

6. MEASUREMENTS FOR TOOTH THICKNESS

		----- Gear 1 -----	Gear 2 --
		DIN 3967 cd25	DIN 3967 cd25
Tooth thickness tolerance			
Tooth thickness allowance (normal section) (mm)	[As.e/i]	-0.070 / -0.110	-0.095 / -0.145
Number of teeth spanned	[k]	5.000	13.000
Base tangent length (no backlash) (mm)	[Wk]	41.862	115.465
Actual base tangent length (span) (mm)	[Wk.e/i]	41.796 / 41.758	115.375 / 115.328
(mm)	[ΔWk.e/i]	-0.066 / -0.103	-0.089 / -0.136
Diameter of measuring circle (mm)	[dMWk.m]	85.832	273.222
Theoretical diameter of ball/pin (mm)	[DM]	5.341	5.060
Effective diameter of ball/pin (mm)	[DMeff]	5.500	5.250
Radial single-ball measurement backlash free (mm)	[MrK]	46.292	139.791
Radial single-ball measurement (mm)	[MrK.e/i]	46.215 / 46.172	139.668 / 139.603
Diameter of measuring circle (mm)	[dMMr.m]	84.643	272.160
Diametral measurement over two balls without clearance (mm)	[MdK]	92.412	279.583
Diametral two ball measure (mm)	[MdK.e/i]	92.259 / 92.172	279.337 / 279.207
Diametral measurement over pins without clearance (mm)	[MdR]	92.583	279.583

Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	92.431 /	92.343	279.337 /	279.207
Measurement over 2 pins (free) according to AGMA 2002 (mm)	[dk2f.e/i]	92.206 /	92.118	0.000 /	0.000
Measurement over 2 pins (axial) according to AGMA 2002 (mm)	[dk2t.e/i]	92.598 /	92.511	0.000 /	0.000
Measurement over 3 pins (axial) according to AGMA 2002 (mm)	[dk3A.e/i]	92.431 /	92.343	279.337 /	279.207
Chordal tooth thickness (no backlash) (mm)	[sc]	5.368	4.928		
Actual chordal tooth thickness (mm)	[sc.e/i]	5.298 /	5.258	4.833 /	4.783
Reference chordal height from da.m (mm)	[ha]	3.943	3.277		
Tooth thickness (Arc) (mm)	[sn]	5.370	4.928		
(mm)	[sn.e/i]	5.300 /	5.260	4.833 /	4.783
Backlash free center distance (mm)	[aControl.e/i]	178.050	/177.930		
Backlash free center distance, allowances (mm)	[jta]	-0.219 /	-0.339		
dNf.i with aControl (mm)	[dNf0.i]	79.068	266.978		
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	-0.029	0.420		
Tip clearance (mm)	[c0.i(aControl)]	0.542	0.508		
Center distance allowances (mm)	[Aa.e/i]	0.020 /	-0.020		
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.017 /	-0.017		
Radial backlash (mm)	[jrw.e/i]	0.359 /	0.199		
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.300 /	0.166		
Normal backlash (mm)	[jn.e/i]	0.253 /	0.141		
Angle of rotation on input with fixed output.:					
Entire torsional angle (°)	[j.tSys]	0.4127/0.2290			

7. GEAR ACCURACY

		----- GEAR 1 -----	GEAR 2 --
According to ISO 1328-1:1995, ISO 1328-2:1997			
Accuracy grade	[Q]	6	6
Single pitch deviation (μm)	[fptT]	8.50	9.00
Base circle pitch deviation (μm)	[fpbT]	7.90	8.40
Sector pitch deviation over k/8 pitches (μm)	[Fpk/8T]	13.00	20.00
Profile form deviation (μm)	[ffaT]	8.50	9.50
Profile slope deviation (μm)	[fHaT]	7.00	8.00
Total profile deviation (μm)	[FaT]	11.00	13.00
Helix form deviation (μm)	[ffbT]	10.00	10.00
Helix slope deviation (μm)	[fHbT]	10.00	10.00
Total helix deviation (μm)	[FbT]	14.00	15.00
Total cumulative pitch deviation (μm)	[FpT]	27.00	35.00
Runout (μm)	[FrT]	21.00	28.00
Single flank composite, total (μm)	[FisT]	37.00	46.00
Single flank composite, tooth-to-tooth (μm)	[fisT]	10.00	11.00
Radial composite, total (μm)	[FidT]	36.00	43.00
Radial composite, tooth-to-tooth (μm)	[fidT]	14.00	15.00

Axis alignment tolerances (recommendation acc. to ISO TR 10064-3:1996, Quality)

6)

Maximum value for deviation error of axis (μm)	[fSigbet]	15.41 (Fb= 15.00)
Maximum value for inclination error of axes (μm)	[fSigdel]	30.82

8. ADDITIONAL DATA

Maximal possible center distance (eps_a=1.0)	[aMAX]	179.861	
Mass (kg)	[m]	2.431	24.795
Total mass (kg)	[m]	27.226	
Moment of inertia (system with reference to the drive): calculation without consideration of the exact tooth shape			
single gears ((da+df)/2...di) (kg*m²)	[TraeghMom]	0.00213	0.22804
System ((da+df)/2...di) (kg*m²)	[TraeghMom]	0.02333	
Torsional stiffness at driving gear with fixed driven gear:			
Torsional stiffness (MNm/rad)	[cr]	0.869	
Torsion when subjected to nominal torque (°)	[delcr]	0.003	
Mean coeff. of friction (acc. Niemann)	[mum]	0.067	
Wear sliding coef. by Niemann	[zetw]	0.602	
Gear power loss (kW)	[PVZ]	0.022	
(Meshing efficiency (%))	[etaz]	99.273	
Sound pressure level (according to Masuda, without contact analysis)			
	[dB(A)]	50.6	

9. MODIFICATIONS AND TOOTH FORM DEFINITION

Data for the tooth form calculation :

Data not available.

Please run the calculation in the "Tooth form" tab and open the main report again.

10. SERVICE LIFE, DAMAGE

Required safety for tooth root	[SFmin]	1.40
Required safety for tooth flank	[SHmin]	1.00

Service life (calculated with required safeties):

System service life (h)	[Hatt]	19271
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Tooth root service life (h)	[HFatt]	1e+006	1e+006
Tooth flank service life (h)	[HHatt]	1.927e+004	6.321e+004

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

Damage calculated on the basis of the required service life [H] (31000.0 h)

F1%	F2%	H1%	H2%
0.00	0.0000	160.8663	49.0446

Damage calculated on basis of system service life [Hatt] (19270.7 h)

F1%	F2%	H1%	H2%
0.00	0.0000	100.0000	30.4878

Calculation of the factors required to define reliability R(t) according to B. Bertsche with Weibull distribution; t in (h):

$$R(t) = 100 * \exp(-((t*fac - t_0)/(T - t_0))^b) \%$$

Gear		fac	b	t0	T	R(H)%
1	Tooth root	45000	1.7	9.654e+029	1.484e+030	100.00
1	Tooth flank	45000	1.3	7.817e+008	3.724e+009	87.79
2	Tooth root	13720	1.7	9.654e+029	1.484e+030	100.00
2	Tooth flank	13720	1.3	7.817e+008	3.724e+009	100.00

Reliability of the configuration for required service life (%) 87.79 (Bertsche)

REMARKS:

- Specifications with [e/i] imply: Maximum [e] and Minimal value [i] with consideration of all tolerances
- Specifications with [m] imply: Mean value within tolerance
- For the backlash tolerance, the center distance tolerances and the tooth thickness deviation are taken into account. Shown is the maximal and the minimal backlash corresponding the largest resp. the smallest allowances
- The calculation is done for the operating pitch circle.
- Calculation of Zbet according Corrigendum 1 ISO 6336-2:2008 with $Z_{bet} = 1/(\cos(\beta))^0.5$
- Details of calculation method:
 - cg according to method B
 - KV according to method B
 - KHb, KFb according method C
 - fma following equation (64), fsh following (57/58), Fbx following (52/53/57)
 - KHa, KFa according to method B
- The logarithmically interpolated value taken from the values for the fatigue strength and the static strength, based on the number of load cycles, is used for coefficients ZL, ZV, ZR, ZW, ZX, YdreIT, YRrelT and YX..

End of Report

lines: 534
